Week 7, October 21, 2026

Main topics

- Bayesian Inverse Problems
- Iterative methods for solving inverse problems

Material
From last week: Statistical and Computational Inverse Problems by Jari Kaipio and Erkki Somersalo, sections 3-3.2.1 (pp 49-58) and 3.4 (pp 72-79), see . Spend in addition 20 minutes with the CUQI-Book.
New Material: P. C. Hansen, *Discrete Inverse Problems: Insight and Algorithms*, Chapter 6 supplemented by Kirsch, section 2.6.

Program

13:00-13:45 Midterm evaluation and MP1.
Inverse problems with Bayes formula: Prior, likelihood and posterior

14:00-16:00 Exercises (see below)

16:15-17:00 Outlook: Iterative methods for inverse problems

Exercises

1. Spend 10 minutes looking at the feedback to MP1. I will then discuss shortcomings with the groups individually.
2. Go through Probably the simplest BIP in the world (the short story). You can run the Python code in browser if you press the rocket and choose Live Code.
3. Assume that X and E are independent Gaussian random variables with prior and noise models

$$X \sim \mathcal{N}(0, \gamma^2 I),$$
$$E \sim \mathcal{N}(0, \sigma^2 I).$$

Bayes' formula then shows that the posterior is Gaussian with density

$$\pi_{\text{post}}(x) = \pi(x|y_{\text{obs}}) = \frac{\pi(y_{\text{obs}}|x)\pi_{\text{pr}}(x)}{\pi(y)} \propto \exp\left(-\frac{1}{2}(x-\overline{x})^T \Gamma_{\text{post}}^{-1}(x-\overline{x})\right).$$

Here

$$\Gamma_{\text{post}} = \gamma^2 - \gamma^4 A^T(\gamma^2 AA^T + \sigma^2)^{-1}A,$$
$$\overline{x} = \gamma^2 A^T(\gamma^2 AA^T + \sigma^2)^{-1}y_{\text{obs}}.$$

- Show using the SVD of A that $\overline{x} = R_\alpha y_{\text{obs}}$ with $\alpha = \sigma^2/\gamma^2$.
- Interpret that higher noise variance requires stronger regularization, and that lower prior variance requires higher regularization. What is your intuition?
- Show that the posterior (in the sense of positive definite matrices) satisfies $\Gamma_{\text{post}} < \gamma^2$. Interpret, that "measurements decrease the uncertainty."

4. We can simulate a Bayesian inverse problem by starting from a deterministic x^*, calculating $y^* = Ax^*$, and then sample the noise e iid from a Gaussian $e \sim \mathcal{N}(0, \gamma^2 I)$, before computing $x_\alpha = R_\alpha(y^* + e)$.
   Use the Abel problem with $N = 256$, $\nu = 1.5$, and the functions $x(s) = s^\nu$, $y(t) = t^{\nu+1/2}\beta(1+\nu, 1/2)$. Let the noise and prior variance be defined by $\gamma = 0.02$ and $\sigma = 0.005$, respectively, and define $\alpha = \sigma^2/\gamma^2$. Repeat, say 100 times, the experiment of sampling the noise, solving the Tikhonov regularized problem $x_\alpha = R_\alpha(y^* + e)$. Plot the obtained distibution of solutions. Compute the sample mean.

5. Show that in the above scheme with $e \sim \mathcal{N}(0, \gamma^2 I)$, you sample from the distribution $R_\alpha(y^* + e) \sim \mathcal{N}(R_\alpha y^*, \Gamma),\ \Gamma = \sigma^2 R_\alpha R_\alpha^T$. Please help me understand that this is the right porterior, $\Gamma = \Gamma_{\text{post}} = \gamma^2(I - R_\alpha A)\ldots$

6. Repeat the above experiment with different noise models (e.g. uniform noise, other mean values etc). Are you sampling from the posterior distribution?

7. Take a look at chapter 3 in the online CUQIpy book; go back to the previous notebooks as you need.